

...sets the width of the line drawn to 3 pixels. The line of code:

```
turnright 10
```

...tells the turtle to turn 10 degrees to the right. At the end of the program, I have added the line of code:

```
go 1000, 1000
```

...to force the turtle to run out of the canvas so that the final image we obtain is not hindered by the presence of the turtle in the middle. Press the 'Run' button and you will see the output on the canvas as shown in Figure 5.

Now it is time to draw my personal favourite, a fractal called the Sierpinski triangle, named after the Polish mathematician Waclaw Sierpiński. The pretty standard code to draw this fractal is taken from the KDE UserBase Wiki and all I have done is add colours to the lines drawn. The code is given below.

```
learn sierp $1 {
  $c = 0
  if $1 > 2 {
    repeat 3 {
      sierp $1/2
      if $c == 0 {
        pencolor 255, 0, 0
      }
      if $c == 1 {
        pencolor 0, 255, 0
      }
      if $c == 2 {
        pencolor 0, 0, 255
      }
      $c = mod ($c + 1), 3
      forward $1
      turnleft 120
    }
  }
}
reset
cansize 600, 600
```

```
go 50, 450
turnright 90
sierp 500
go 1000, 1000
```

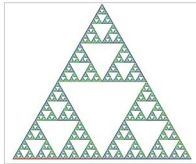


Figure 6: Sierpinski triangle with K-Turtle

Here, a command called *learn* is used to create our own command called *sierp*. This is similar to defining user-defined functions in traditional programming languages like C or Java. This function *sierp* is called

recursively to draw the fractal. If you observe the program carefully, you will see that two variables 'l' and 'c' are used for processing. Operations like addition and modulus are performed on these variables. The mathematical implications of this fractal are pretty amazing but the aesthetic pleasure offered by the image of the fractal is even more so. Figure 6 shows the output of the program on execution.

This is just a basic introduction, yet sufficient enough to get people interested in the language, I believe. In a theoretical sense, K-Turtle is a complete programming language which is as powerful as general-purpose programming languages like C or Java. But no one uses K-Turtle for actual programming; instead, it is solely used as a tool to teach programming to children. As Nelson Mandela said, "Education is the most powerful weapon which you can use to change the world." So, what other programming language can be more important than the one that will inspire and develop young programmers. **END** 🐢

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